

CycleGAN: Horse to Zebra Unpaired Image-to-Image Translation

with Proposed Perceptual Cycle Consistency Loss

Deep Learning for Perception
Final Semester Project Report

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May 9, 2026

1. Task Definition

This project addresses the problem of unpaired image-to-image translation, a computer vision task where the goal is to learn a mapping between two visual domains without any paired training examples. Specifically, the task involves translating photographs of horses into photographs that resemble zebras, and vice versa.

The framework used to solve this task is CycleGAN, proposed by Zhu et al. (2017). CycleGAN learns two translation functions simultaneously: one that maps horse images to the zebra domain, and another that maps zebra images back to the horse domain. A cycle consistency constraint is enforced so that translating an image from one domain to the other and back again recovers the original image. This constraint allows the model to train without any paired correspondence between the two domains.

Beyond replicating the standard CycleGAN baseline, this project also proposes a modification to the cycle consistency loss. The standard approach penalises pixel-level differences between the original and reconstructed images, which can lead to blurry outputs. The proposed modification replaces this pixel-level penalty with a perceptual loss computed in the feature space of a pretrained VGG-16 network. The hypothesis is that comparing images at the level of learned visual features produces sharper, more structurally faithful reconstructions.

2. Dataset Description

The Horse2Zebra dataset, introduced alongside the CycleGAN paper (Zhu et al., 2017), is used for this project. The dataset consists of RGB colour photographs sourced from the ImageNet horse synset (n02381460) and zebra synset (n02391049). All images are in JPEG format.

The dataset is split into four directories. The table below summarises the composition of each split:

Directory	Domain	Number of Images
trainA	Horses (training)	1,067
trainB	Zebras (training)	1,334
testA	Horses (test)	120
testB	Zebras (test)	140

The dataset does not contain any labels in the conventional sense. There are no bounding boxes, segmentation masks, or class annotations. The only supervision available is domain

membership, meaning each image is known to belong to either the horse domain or the zebra domain. The task is therefore fully unsupervised with respect to image content.

Importantly, the training images are unpaired. There is no correspondence between any horse image in trainA and any zebra image in trainB. The model must discover the visual differences between the two domains entirely from their respective distributions.

It is worth noting that the zebra training set is approximately 25 percent larger than the horse training set. This imbalance partially contributes to the observed asymmetry in translation quality, where the horse to zebra direction tends to produce better results than the reverse.

3. Data Pre-processing

3.1 Training Pipeline

Each training image undergoes the following sequence of transformations before being passed to the network:

- Resize: Each image is resized to 286 by 286 pixels using bicubic interpolation. This is intentionally larger than the final training resolution to allow for spatial jitter in the next step.
- Random Crop: A 256 by 256 patch is randomly cropped from the 286 by 286 image. This introduces positional augmentation and prevents the network from overfitting to edge locations.
- Random Horizontal Flip: The image is flipped horizontally with a probability of 0.5. This effectively doubles the number of training examples and improves the model's left-right invariance.
- Normalisation: Pixel values, originally in the range 0 to 1, are normalised to the range negative 1 to positive 1 using a mean of 0.5 and a standard deviation of 0.5 for each colour channel. This range matches the output range of the generator's Tanh activation.

3.2 Test and Inference Pipeline

During evaluation and inference, augmentation is disabled to ensure deterministic results. Each image is resized directly to 256 by 256 pixels using bicubic interpolation, followed by the same normalisation step as training. No random cropping or flipping is applied.

No additional pre-processing steps such as noise reduction, histogram equalisation, or colour jitter are applied at any stage. Since the domain difference between horses and zebras is primarily one of texture rather than illumination or geometry, aggressive colour

augmentation would interfere with the style signal the model needs to learn.

The data loading class samples images from Domain A and Domain B independently within each batch. No attempt is made to create or enforce any pairing between horse and zebra images during training.

4. Network Architecture

The overall system consists of four neural networks trained simultaneously: two generators and two discriminators. The generators learn the translation mappings between domains, while the discriminators assess whether generated images are realistic.

4.1 Generator

The generator follows the ResNet-based encoder-decoder architecture proposed by Johnson et al. (2016), adapted for use in CycleGAN. The network consists of three stages: a downsampling encoder, a sequence of residual blocks that transform the feature representation, and an upsampling decoder that reconstructs the image at full resolution.

The encoder begins with a large-kernel convolutional layer that processes the full-resolution input and produces 64 feature maps. This is followed by two strided convolutional layers that progressively halve the spatial resolution while doubling the number of channels, arriving at a compact representation of 256 channels at one quarter of the original resolution.

The bottleneck consists of nine residual blocks, each containing two convolutional layers with a skip connection. This depth is recommended by Zhu et al. for images of 256 by 256 resolution. The residual connections allow gradient flow during training and help the generator preserve content structure while modifying style.

The decoder mirrors the encoder using two transposed convolutional layers to progressively restore spatial resolution, followed by a final convolutional layer that maps to three output channels. A Tanh activation at the output confines pixel values to the range negative 1 to positive 1.

Throughout the generator, instance normalisation is used in place of batch normalisation. Instance normalisation normalises each image independently, which is important for style-transfer tasks where per-image statistics are meaningful. Reflection padding is used throughout to reduce border artifacts that commonly arise with zero-padding. Each generator contains approximately 11.4 million trainable parameters.

4.2 Discriminator

The discriminator uses the PatchGAN architecture (Isola et al., 2017), which classifies overlapping local patches of the input image as real or fake rather than classifying the entire image with a single scalar output. This design encourages the generator to produce realistic high-frequency textures, which is particularly important for the stripe patterns in the horse to zebra translation.

The network applies five convolutional layers with progressively increasing channel depth. LeakyReLU activations with a slope of 0.2 are used throughout. Instance normalisation is applied to all layers except the first, which is left unnormalised to preserve low-level colour and texture statistics. The final layer produces a patch-level map of real-versus-fake scores with no sigmoid activation, as the loss function operates directly on the raw outputs. Each discriminator has an effective receptive field of 70 by 70 pixels and contains approximately 2.8 million parameters.

The full system comprises two generators (one for each translation direction) and two discriminators (one for each domain), giving a total of approximately 28.4 million trainable parameters.

4.3 Proposed Modification: Perceptual Cycle Loss

The proposed variant of the architecture introduces a change only to the loss function used for cycle consistency training. The generator and discriminator network structures are identical to the baseline. The modification involves adding a frozen, pretrained VGG-16 network that serves as a fixed feature extractor. This network is never updated during training. Features are extracted from four intermediate layers of VGG-16, corresponding to progressively deeper levels of abstraction. The cycle consistency penalty is then computed in this feature space rather than directly on pixel values, as described in Section 5.

5. Loss Function

The total training objective combines three types of loss, each targeting a different aspect of the translation quality.

5.1 Adversarial Loss

The adversarial loss encourages the generator to produce images that are indistinguishable from real images in the target domain. This project uses the Least Squares GAN formulation (Mao et al., 2017) rather than the original binary cross-entropy loss. The least squares objective is more stable during training and tends to produce sharper results because it penalises samples that lie far from the decision boundary, rather than those that merely fool the discriminator with high confidence.

The discriminator is trained to output a value of 1 for real images and 0 for generated images. The generator is trained to make the discriminator assign a value of 1 to its outputs. An adversarial loss is applied independently for both translation directions.

5.2 Cycle Consistency Loss

The cycle consistency loss addresses the core challenge of unpaired training. Without a direct pixel-level supervision signal, the adversarial loss alone cannot constrain which horse image maps to which zebra image, leading to mode collapse. Cycle consistency enforces that if a horse image is translated to the zebra domain and then back to the horse domain, the result should closely match the original horse image, and vice versa.

In the baseline model, this constraint is enforced with an L1 penalty in pixel space. The L1 loss is weighted by a factor of 10, following the ablation study in Zhu et al. (2017). In the proposed modification, this pixel-space penalty is replaced by a perceptual loss. The reconstructed image and the original image are passed through a frozen VGG-16 network, and the difference between their feature maps is minimised at four intermediate layers (relu1_2, relu2_2, relu3_2, and relu4_2). The perceptual loss is applied with the same overall weight of 10, with equal contribution from each VGG layer. The rationale is that feature-space distances better capture perceptual and structural similarity than raw pixel differences, leading to sharper reconstructions.

5.3 Identity Loss

The identity loss is an auxiliary regularisation term that encourages the generator to act as a near-identity mapping when given an image that already belongs to the target domain. For example, passing a zebra image through the horse-to-zebra generator should return an image close to the original zebra. This helps preserve the colour characteristics of the input and prevents unnecessary changes to images that do not need to be translated. The identity loss is computed using an L1 penalty in pixel space in both the baseline and modified models, with a weight of 5 (equal to half the cycle consistency weight).

5.4 Summary of Loss Weights

The table below summarises the three loss terms and their weights in both the baseline and proposed modified models:

Loss Term	Weight	Baseline	Modified
Adversarial loss	1.0	L1 pixel (LSGAN)	L1 pixel (LSGAN)
Cycle consistency loss	10	L1 in pixel space	MSE in VGG-16 feature space

Loss Term	Weight	Baseline	Modified
Identity loss	5	L1 in pixel space	L1 in pixel space

6. Hyperparameters

All core hyperparameters were adopted directly from the original CycleGAN paper (Zhu et al., 2017) to ensure a faithful and reproducible baseline. Deviations were made only where hardware constraints required it, as noted in the table below.

Parameter	Value	Selection Rationale
Image size	256 x 256	Standard benchmark resolution for Horse2Zebra
Batch size	4	Constrained by available GPU memory (16 GB VRAM)
Learning rate	0.0002	Adopted directly from Zhu et al. (2017)
Optimizer	Adam	Standard choice for adversarial training
Adam beta 1	0.5	Recommended for GAN stability by Radford et al. (2016)
Adam beta 2	0.999	Default Adam value
Total epochs	200	Standard CycleGAN training schedule
LR decay start epoch	100	Linear decay from epoch 100 to 200, reaching zero
Residual blocks	9	Recommended by Zhu et al. for 256x256 inputs
Cycle loss weight	10	Selected via ablation study in Zhu et al. (2017)
Identity loss weight	5	Set to half the cycle loss weight, per Zhu et al.
Replay buffer size	50 images	Following recommendation of Shrivastava et al. (2017)
Mixed precision (AMP)	Enabled	Used for training efficiency on GPU
Data loading workers	8	Matched to the number of CPU cores available

The learning rate is held constant at 0.0002 for the first 100 epochs and then linearly decayed to zero over the remaining 100 epochs. This schedule allows the model to explore the parameter space freely in the early stages and then refine details as training progresses.

The replay buffer stores up to 50 previously generated images and is used to update the discriminator. At each discriminator update, either the current generated image or a randomly selected image from the buffer is used, each with 50 percent probability. This prevents the discriminator from adapting too rapidly to the current state of the generator, which improves training stability.

7. SOTA Comparison

7.1 Quantitative Comparison

The standard metric for evaluating image translation quality is the Frechet Inception Distance (FID). FID measures the statistical distance between the distribution of generated images and the distribution of real images, as estimated in the feature space of a pretrained Inception-v3 network. Lower FID values indicate better quality.

The table below reports published FID scores on the Horse2Zebra benchmark for several representative methods, along with the status of our own results:

Method	FID H to Z (lower is better)	FID Z to H (lower is better)	Year
CycleGAN (Zhu et al.)	77.2	138.1	2017
UNIT (Liu et al.)	96.1	145.3	2017
MUNIT (Huang et al.)	74.8	136.8	2018
CUT (Park et al.)	45.5	85.9	2020
Our Baseline (Epoch 200)	66.28	144.08	2026
Our Modified (relu3_2, Epoch 90)	230.08	235.39	2026

Quantitative evaluation of both models was carried out on the standard Horse2Zebra test set (120 horse images in testA, 140 zebra images in testB). Our baseline model, evaluated using the epoch 200 checkpoint, achieves an FID of 66.28 on Horse to Zebra and 144.08 on Zebra to Horse. Notably, the Horse to Zebra FID of 66.28 improves upon the published CycleGAN result of 77.2, which reflects the benefit of completing the full 200-epoch training schedule including the linear learning rate decay phase (epochs 100 to 200). The Zebra to Horse FID of 144.08 is slightly above the published 138.1, consistent with the known difficulty of this direction documented in the original paper. Our modified model (relu3_2 perceptual cycle loss, epoch 90 checkpoint) achieves FID values of 230.08 and 235.39 for the two directions respectively. These values reflect the model being only partway through training, with the perceptual loss landscape requiring additional epochs to converge.

In addition to FID, cycle consistency was measured using the Structural Similarity Index (SSIM) between original images and their cycle-reconstructed counterparts. For the baseline model, the Horse cycle SSIM is 0.8391 and the Zebra cycle SSIM is 0.8731, giving a mean SSIM of 0.8574. For the modified model, the Horse cycle SSIM is 0.9394 and the Zebra cycle SSIM is 0.9487, giving a mean SSIM of 0.9444. The notably higher SSIM of

the modified model indicates that the perceptual cycle loss enforces stronger structural consistency in round-trip reconstruction, even though the model has not yet converged in terms of FID. These scores confirm that both models preserve structural content through the full translation round-trip.

7.2 Qualitative Comparison

Qualitative results for the baseline model are available from the training run, with sample outputs saved at every 10th epoch from epoch 10 to epoch 200. Qualitative results for the modified model are available from epoch 10 to epoch 90. The following observations summarise the translation quality:

- Horse to Zebra: By epoch 80, the generator produces convincing stripe patterns that follow the body contours of the horse. By epoch 200, the texture is well-defined and perceptually plausible. Background regions are largely unaffected. The results are qualitatively comparable to the published examples in the original CycleGAN paper.
- Zebra to Horse: Stripe removal is partially successful, but the synthesised coat texture can appear inconsistent or overly smooth in some cases. This asymmetry between the two translation directions is a known limitation of the CycleGAN framework and is documented in the original paper. Adding stripes to a horse is an additive operation, whereas removing zebra stripes requires synthesising plausible smooth texture beneath a strong high-frequency pattern, which is inherently more difficult.
- Training stability: Generator loss decreased steadily from approximately 3.6 at epoch 71 to approximately 2.7 at epoch 150. Discriminator loss remained in the range of 0.17 to 0.54 throughout, indicating stable adversarial training without collapse or oscillation.
- Metric consistency: The directional asymmetry observed qualitatively is confirmed by both metrics. FID is better in the Horse to Zebra direction (66.28) than Zebra to Horse (144.08), while SSIM is slightly higher for the Zebra cycle (0.8731) than the Horse cycle (0.8391). The stronger SSIM for Zebra cycle despite the weaker FID for that direction reflects the different aspects each metric captures: FID measures the realism of the generated images, whereas SSIM measures how faithfully the original is reconstructed after the full translation round-trip.

7.3 Positioning Against the State of the Art

Our fully trained baseline achieves an FID of 66.28 on Horse to Zebra at epoch 200, which improves upon the published CycleGAN result of 77.2 and places our implementation between MUNIT (74.8) and CUT (45.5) on this benchmark. This is a positive result, demonstrating that the reimplementations are faithful to the original paper and that completing

the full 200-epoch training schedule — including the linear learning rate decay phase — meaningfully improves generation quality. On the Zebra to Horse direction, our FID of 144.08 is slightly above the published 138.1, consistent with the known difficulty of this direction. The current strongest published result on this benchmark is CUT (Park et al., 2020), which achieves an FID of 45.5 using a contrastive learning objective rather than cycle consistency. The proposed perceptual loss modification was trained to epoch 90 and currently shows high FID (230.08) but substantially higher SSIM (0.9444 vs 0.8574 for the baseline). The improved cycle consistency under perceptual loss is an encouraging early signal; full convergence with extended training is needed to draw conclusions about FID.

8. Remaining Work and Limitations

Due to limited GPU cluster access and file corruption during checkpoint transfer, some planned components could not be fully executed. The table below summarises the status of each component:

Component	Status	Notes
Baseline model code	Complete	—
Modified perceptual loss model code	Complete	—
Baseline training, epochs 1 to 200	Complete	—
Inference web application (Gradio)	Complete	—
Modified model training (relu3_2)	Partial — epoch 90	Resource and time constraints
Modified model checkpoints	Partial	Only epochs 10, 20, 80, 90 retained (others corrupted)
Quantitative evaluation of baseline (FID, SSIM)	Complete	—
Quantitative evaluation of modified model	Complete	Epoch 90 checkpoint used
VGG layer ablation study	Not run	Insufficient modified checkpoints available
Training loss logs (baseline)	Partial	Epochs 71 to 200 available; earlier logs not retained
Training loss logs (modified)	Missing	Logs not copied before losing cluster access

8.1 Impact of Missing and Incomplete Components

The baseline model was trained to completion at 200 epochs, achieving an FID of 66.28 on Horse to Zebra and 144.08 on Zebra to Horse. The Horse to Zebra result improves upon the published CycleGAN score of 77.2, which is a strong positive outcome for the reimplementation.

The modified perceptual loss model was trained to epoch 90 out of a planned 200 epochs. At this stage, the FID values are high (230.08 and 235.39) as the model has not yet converged, but the mean SSIM of 0.9444 substantially exceeds the baseline mean SSIM of 0.8574. This indicates that the perceptual cycle loss enforces stronger structural consistency in round-trip reconstruction, which is the core motivation of the proposed modification. The hypothesis that perceptual loss improves structural fidelity is supported by these early results; the question of whether it also improves FID at convergence remains open.

The VGG layer ablation study could not be completed because intermediate checkpoints for the modified model (epochs 30 to 70, and 100 onwards) were corrupted during file transfer and are no longer available. Only epoch 10, 20, 80, and 90 checkpoints survive. Loss logs for the modified model were not retained before cluster access was lost.

8.2 Planned Future Work

- Continue training the modified perceptual loss model from epoch 90 to 200 epochs and run a full quantitative and qualitative comparison against the baseline to determine whether perceptual cycle loss also improves FID at convergence.
- Execute the VGG layer ablation study by training separate modified models with cycle consistency measured at each of `relu1_2`, `relu2_2`, `relu3_2`, and `relu4_2` independently, to determine which level of abstraction is most effective for cycle consistency.
- Investigate methods to reduce the Zebra to Horse quality gap, such as applying asymmetric loss weights that impose stronger cycle consistency on the harder direction.
- Extend the comparison to include CUT (Park et al., 2020) as a stronger modern baseline.

References

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